

Chapter 24

Lecture 24: Stone–Čech compactification. Complete metric spaces.

Theorem 24.0.1. [Recall: Stone–Čech compactification] *Let X be a completely regular space. Then there exists a compactification Y of X such that every bounded continuous function $f : X \rightarrow \mathbb{R}$ extends uniquely to a continuous function $\bar{f} : Y \rightarrow \mathbb{R}$. (Y is called the Stone–Čech compactification of X)*

In the last lecture, we have shown the existence part of the above theorem. The uniqueness part is a consequence of the following lemma.

Lemma 24.0.1. *Let $A \subseteq X$. Let $f : A \rightarrow Z$ be continuous and Z be Hausdorff. Then there exists at most one extension of f to a continuous function $g : \bar{A} \rightarrow Z, g|_A = f$.*

Proof

Suppose that $g, g' : \bar{A} \rightarrow Z$ are two different extensions of f . Then there exists $x \in \bar{A}$ such that $g(x) \neq g'(x)$. Since Z is Hausdorff, there exist disjoint open sets U, U' in Z such that $g(x) \in U$ and $g'(x) \in U'$. Choose a neighborhood V of x such that $g(V) \subseteq U$ and $g'(V) \subseteq U'$. Now V intersects A because $x \in \bar{A}$. Take $y \in V \cap A$. Then $g(y) = f(y) = g'(y)$, contradicting the fact that U and U' are disjoint. ■

Theorem 24.0.2. *Let X be completely regular and Y be a compactification of X such that Y satisfies the Stone–Čech property 24.0.1. Then any continuous map $f : X \rightarrow C$ with C being a compact Hausdorff space can be extended uniquely to a continuous map $g : Y \rightarrow C$.*

Proof

C is compact Hausdorff, so it is normal thus is completely regular. So C can be imbedded in a cube $[0, 1]^J$ for some index set J . Since C is homeomorphic to its image in $[0, 1]^J$, we may as well assume C is a subspace of $[0, 1]^J$. Then each component function $f_\alpha = \pi_\alpha \circ f : X \rightarrow [0, 1]$ is a bounded continuous function on X . By the Stone–Čech property, f_α extends to a continuous function $\bar{f}_\alpha : Y \rightarrow [0, 1]$. Define $\bar{f} : Y \rightarrow [0, 1]^J$ by $\bar{f}(y) = (\bar{f}_\alpha(y))_{\alpha \in J}$. Then $\bar{f}$ is continuous and extends f . Now in fact $\bar{f}$ maps Y into C because

$$\bar{f}(Y) = \bar{f}(\bar{X}) \subseteq \overline{\bar{f}(X)} = \overline{f(X)} \subseteq \bar{C} = C. \quad (24.1)$$

where $Y = \bar{X}$ because X is dense in Y , $\bar{f}(\bar{X}) \subseteq \overline{\bar{f}(X)}$ because $\bar{f}$ is continuous, $\overline{\bar{f}(X)} = \overline{f(X)}$ because $\bar{f}$ extends f , and $\bar{C} = C$ because C is closed as a compact subspace of a Hausdorff space. So $\bar{f}$ is a desired extension of f . The uniqueness of the extension follows from the previous lemma. ■

Theorem 24.0.3. *Let Y_1, Y_2 be compactifications of a completely regular space X . Suppose both Y_1 and Y_2 satisfy the Stone–Čech property. Then there exist homeomorphisms $h : Y_1 \rightarrow Y_2$.*

Proof

Consider the inclusion mapping $j_2 : X \rightarrow Y_2$. It is a continuous map of X into the compact Hausdorff space Y_2 . Because Y_1 satisfies the Stone–Čech property, by the previous theorem, j_2 extends to a continuous map $f_2 : Y_1 \rightarrow Y_2$. Similarly, the inclusion mapping $j_1 : X \rightarrow Y_1$ extends to a continuous map $f_1 : Y_2 \rightarrow Y_1$.

The composite $f_1 \circ f_2 : Y_1 \rightarrow Y_1$ has the property that $f_1 \circ f_2(x) = x$ for all $x \in X$. Therefore, $f_1 \circ f_2$ is a continuous extension of the identity map on X . But the identity map of Y_1 is also a continuous extension of the

identity map on X . By the uniqueness of the extension 24.0.1, we have $f_1 \circ f_2 = id_{Y_1}$. Similarly, $f_2 \circ f_1 = id_{Y_2}$. So f_1 and f_2 are homeomorphisms. ■

24.1 Complete Metric Spaces

Definition 24.1.1. Let (X, d) be a metric space. A sequence $\{x_n\} \subseteq X$ is called a *Cauchy sequence* if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, $d(x_n, x_m) < \varepsilon$.

Property 24.1.1. Every convergent sequence is a Cauchy sequence.

Proof Easy exercise.

Definition 24.1.2. A metric space (X, d) is called *complete* if every Cauchy sequence in X converges to a limit in X .

Example 24.1.1 (non-example). $(0, 1)$ with the usual metric. Then the sequence $x_n = \frac{1}{n}$ is a Cauchy sequence that does not converge in $(0, 1)$.

Remark This example also shows that completeness is not a topological property. The subspace $(0, 1)$ of $\mathbb{R}$ is homeomorphic to $\mathbb{R}$, but $(0, 1)$ is not complete while $\mathbb{R}$ is complete.

Example 24.1.2 (non-example). $\mathbb{Q}$ with the usual metric is not complete. For example, the sequence defined by the decimal approximations of $\sqrt{2}$ is a Cauchy sequence in $\mathbb{Q}$ that does not converge to a rational number.

Property 24.1.2. If $A \subseteq X$ is closed and (X, d) is complete, then A is complete in the restricted metric $d|_A$.

Proof Easy exercise.

Property 24.1.3. Let (X, d) be a metric space. Take $\bar{d} = \min\{d, 1\}$. Then (X, d) is complete implies $(X, \bar{d})$ is complete, and conversely.

Proof

A sequence $\{x_n\}$ is Cauchy in d if and only if it is Cauchy in $\bar{d}$. And a sequence $\{x_n\}$ converges under d if and only if it converges under $\bar{d}$. ■

Lemma 24.1.1. A metric space X is complete if and only if every Cauchy sequence has a convergent subsequence.

Proof

“ $\Rightarrow$ ”: Trivial.

“ $\Leftarrow$ ”:

Let $\{x_n\}$ be a Cauchy sequence in (X, d) . We show that if $\{x_{n_k}\}$ is a subsequence of $\{x_n\}$ that converges to some $x \in X$, then $\{x_n\}$ converges to x .

Given $\epsilon > 0$, there exists $N_1 \in \mathbb{N}$ such that for all $m, n \geq N_1$, $d(x_n, x_m) < \frac{\epsilon}{2}$. Since $\{x_{n_k}\}$ converges to x , there exists $N_2 \in \mathbb{N}$ such that for all $n_k \geq N_2$, $d(x_{n_k}, x) < \frac{\epsilon}{2}$. Let $N = \max\{N_1, N_2\}$. Then for all $n, n_k \geq N$, we have

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \quad (24.2)$$

Theorem 24.1.1. $\mathbb{R}^k$ is complete in d_1, d_2, d_∞ .

Proof

We show this for d_∞ .

Let $\{x_n\} \subseteq \mathbb{R}^k$ be a Cauchy sequence in d_∞ . Then $\{x_n\}$ is bounded, i.e., there exists $M > 0$ such that for all n , $d_\infty(x_n, 0) < M$. So such sequence lies in a compact set $[-M, M]^k$. We know in metric spaces, compactness is equivalent to sequential compactness. Thus there exists a convergent subsequence $\{x_{n_k}\}$ that converges to some $x \in [-M, M]^k$. By the lemma, X is complete under d_∞ .

All these metrics are equivalent, so $\mathbb{R}^k$ is complete in d_1, d_2 as well. ■

Lemma 24.1.2. Let $X = \prod_{\alpha \in J} X_\alpha$. Let (x_n) be a sequence in X . Then (x_n) converges to $x \in X$ if and only if for all $\alpha \in J$, the sequence of α -th coordinates $(\pi_\alpha(x_n))$ converges to $\pi_\alpha(x) \in X_\alpha$.

Proof

“ $\Rightarrow$ ”: π_α is continuous for all $\alpha \in J$. Then it preserves convergence.

“ $\Leftarrow$ ”: Suppose $(\pi_\alpha(x_n))$ converges to $\pi_\alpha(x)$ for all $\alpha \in J$. Let U be an open neighborhood of x . Then there exists a finite set $K \subseteq J$ and open sets $U_\alpha \subseteq X_\alpha$ for all $\alpha \in K$ such that $x \in \prod_{\alpha \in K} U_\alpha \times \prod_{\alpha \in J \setminus K} X_\alpha \subseteq U$. Since $(\pi_\alpha(x_n))$ converges to $\pi_\alpha(x)$, for each $\alpha \in K$, there exists $N_\alpha \in \mathbb{N}$ such that for all $n \geq N_\alpha$, $\pi_\alpha(x_n) \in U_\alpha$. Let $N = \max_{\alpha \in K} N_\alpha$. Then for all $n \geq N$, $x_n \in U$. Thus (x_n) converges to x . $\blacksquare$

Theorem 24.1.2. *There is a metric on $\mathbb{R}^\omega$ that makes it complete.*

Proof

Define $\bar{d}(a, b) = \min\{1, |a - b|\}$ for all $a, b \in \mathbb{R}$. Define $D(x, y) = \sup_{i \in \mathbb{N}} \{\frac{\bar{d}(x_i, y_i)}{i}\}$ for all $x = (x_i), y = (y_i) \in \mathbb{R}^\omega$. Then D is a metric on $\mathbb{R}^\omega$ that induces the product topology (Please refer to Lecture 15).

Let $\{x_n\} \subseteq \mathbb{R}^\omega$ be a Cauchy sequence in D . Because

$$\bar{d}(\pi_i(x), \pi_i(y)) \leq i \cdot D(x, y) \quad (24.3)$$

we see that for fixed i , $\{\pi_i(x_n)\}$ is a Cauchy sequence in $\mathbb{R}$ under $\bar{d}$ thus is a Cauchy sequence under d . Then it converges, say, to a_i under d . Then the sequence x_n converges to $a = (a_i) \in \mathbb{R}^\omega$ by the previous lemma. $\blacksquare$

24.2 Uniform Metric

Let (Y, d) be a metric space. Let $\bar{d} = \min\{d, 1\}$. We take the set of functions Y^J . There is a metric $\bar{\rho}$ on Y^J defined by $\bar{\rho}(x, y) = \sup_{\alpha \in J} \{\bar{d}(x_\alpha, y_\alpha)\}$. This is a metric (exercise). It is called the uniform metric.

Example 24.2.1. Let $J = [0, 1]$ and $Y = \mathbb{R}$. Define $\bar{d}(f, g) = \sup_{x \in [0, 1]} \{\min(1, |f(x) - g(x)|)\}$.

Theorem 24.2.1. *If Y is complete with respect to d , then Y^J is complete in the uniform metric $\bar{\rho}$ corresponding to d .*

Proof

Recall that if (Y, d) is complete, so is $(Y, \bar{d})$. Now suppose that $f_1, f_2, \dots$ is a Cauchy sequence in Y^J under $\bar{\rho}$. Given $\alpha \in J$, the fact that

$$\bar{d}(f_n(\alpha), f_m(\alpha)) \leq \bar{\rho}(f_n, f_m) \quad (24.4)$$

for all n, m implies that $\{f_n(\alpha)\}$ is a Cauchy sequence in Y under $\bar{d}$. Then it converges to some point y_α . Let f be defined by $f(\alpha) = y_\alpha$. We assert that f_n converges to f in the uniform metric $\bar{\rho}$.

Given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$, $\bar{\rho}(f_n, f_m) < \frac{\epsilon}{2}$. In particular, given $\alpha \in J$,

$$\bar{d}(f_n(\alpha), f_m(\alpha)) < \frac{\epsilon}{2} \quad (24.5)$$

for all $n, m \geq N$. Letting n and α be fixed and letting $m \rightarrow \infty$ (classical and trivial method in Calculus), we have that

$$\bar{d}(f_n(\alpha), f(\alpha)) \leq \frac{\epsilon}{2} \quad (24.6)$$

This inequality holds for all $\alpha \in J$, provided $n \geq N$. Therefore,

$$\bar{\rho}(f_n, f) = \sup_{\alpha \in J} \{\bar{d}(f_n(\alpha), f(\alpha))\} \leq \frac{\epsilon}{2} < \epsilon \quad (24.7)$$

for all $n \geq N$. So f_n converges to f in the uniform metric $\bar{\rho}$. $\blacksquare$

Remark Recall that the product topology on $\mathbb{R}^J$ is not metrizable if J is uncountable. The topology induced by the uniform metric on $\mathbb{R}^J$ is strictly finer than the product topology so the uniform metric does not metrize the product topology.

The set J here is merely a set, not necessarily a topological space. If J is a topological space, then we have the following result.

Theorem 24.2.2. *Let X be a topological space and let (Y, d) be a metric space. The set $C(X, Y)$ of continuous functions from X to Y is closed in Y^X under the uniform metric. So is the set $B(X, Y)$ of bounded functions. Therefore, if Y is complete, these spaces are complete in the uniform metric.*

Definition 24.2.1. *If (Y, d) is a metric space, one can define another metric on the set $B(X, Y)$ of bounded functions from X to Y by the equation*

$$\rho(f, g) = \sup_{x \in X} \{d(f(x), g(x))\} \quad (24.8)$$

It is easy to see that ρ is well-defined, for the set $f(X) \cup g(X)$ is bounded if both $f(X)$ and $g(X)$ are bounded. The metric ρ is called the sup metric.

Remark There is something.

24.3 Completion

Theorem 24.3.1. *Let (X, d) be a metric space. There is an isometric imbedding of X into a complete metric space.*

Exercise The completion of X is unique up to isometry.